

Geometric Descent Theorems for Pandrosion Flows

Conditional Lyapunov statements and descent conjectures for finite-slope dynamics

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Abstract

This paper formulates Pandrosion as a finite-slope descent process. We define stable probe pairs, introduce a residual energy, prove a conditional finite-slope descent theorem, and state conjectures connecting equal-value avoidance, logarithmic normalization, and gated finite-slope Halley corrections to monotone energy decrease.

Scope and status. This manuscript is written as a theorem-and-conjecture paper inside the Pandrosion research programme. It gives precise definitions, conditional theorems, and conjectures that can be tested by the existing finite-slope engines. The results are structural: they are not trading models and do not rely on empirical market data.

Variational Principles for Finite-Slope Pandrosion Flows

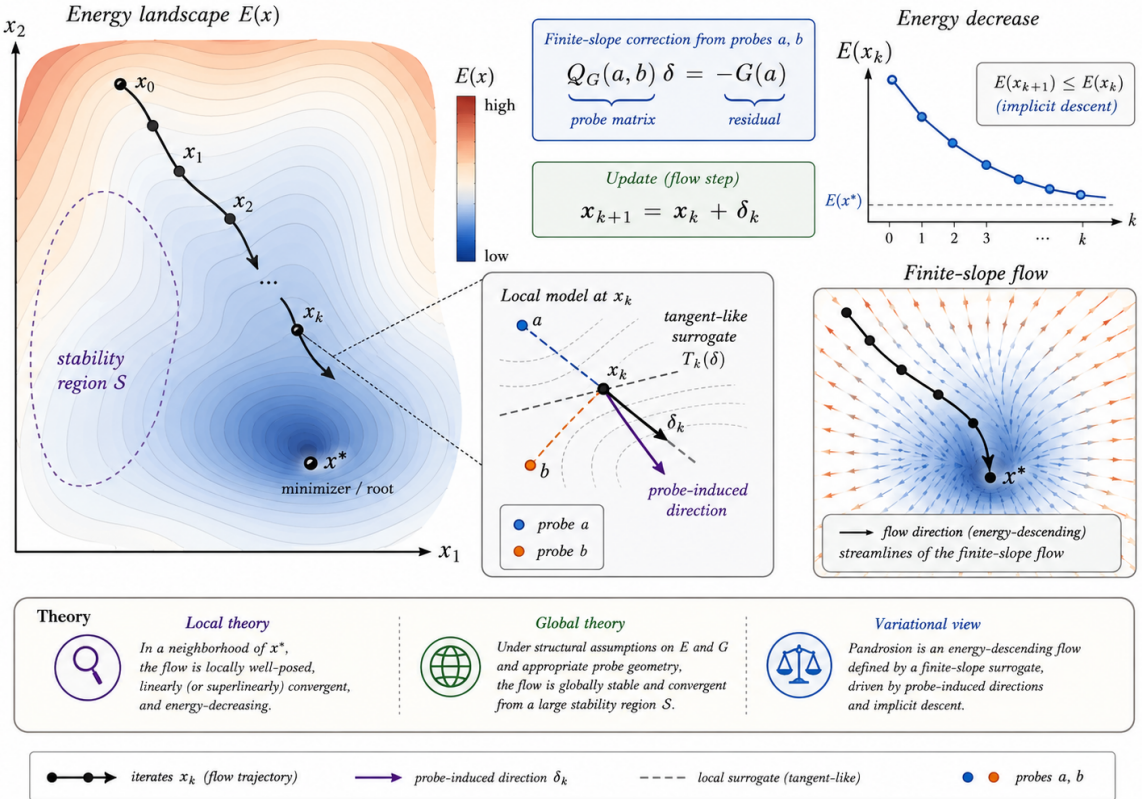


Figure 1: Conceptual overview for this paper. The diagram is intentionally synthetic: it organizes the geometric objects used in the definitions and conjectures.

1 Finite-slope residual geometry

Let $G : U \subset \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a residual map and let $a \in U$ be the current iterate. A Pandrosion step is defined by selecting a probe b and solving

$$Q_G(a, b) \delta = -G(a), \quad a_+ = a + \lambda \delta, \quad (1)$$

where $Q_G(a, b)$ is a finite-slope matrix assembled from probe evaluations and $\lambda \in (0, 1]$ is a line-search or flow-time parameter. The associated residual energy is

$$\mathcal{E}(x) = \frac{1}{2} \|G(x)\|_2^2. \quad (2)$$

The goal of this paper is to formulate conditions under which $\mathcal{E}$ is a Lyapunov function for the Pandrosion flow.

Definition 1.1 (Stable finite-slope pair). *A pair (a, b) is called (μ, K, η) -stable for G if $Q_G(a, b)$ is invertible and*

$$\|Q_G(a, b)^{-1}\| \leq K, \quad (3)$$

$$\Re \langle G(a), Q_G(a, b) Q_G(a, b)^{-1} G(a) \rangle \geq \mu \|G(a)\|^2, \quad (4)$$

$$\frac{\|G(b) - G(a)\|}{\|G(a)\| + \|G(b)\| + 10^{-300}} \geq \eta. \quad (5)$$

The last inequality is the equal-value separation condition; it excludes probes lying close to $G(a) = G(b)$.

2 A descent theorem

The next statement is deliberately conditional. It is not a claim that every probe gives descent; it states what must be verified by the probe engine.

Assumption 2.1 (Finite-slope Taylor defect). *There exists $L > 0$ such that for the Pandrosion direction δ and all $0 \leq \lambda \leq 1$,*

$$\|G(a + \lambda \delta) - G(a) - \lambda Q_G(a, b) \delta\| \leq \frac{L}{2} \lambda^2 \|\delta\|^2. \quad (6)$$

Theorem 2.2 (Geometric finite-slope descent). *Assume that (a, b) is (μ, K, η) -stable and that the finite-slope Taylor defect bound holds. Let δ solve $Q_G(a, b) \delta = -G(a)$. If*

$$0 < \lambda \leq \min \left(1, \frac{\mu}{LK^2 \|G(a)\| + 10^{-300}} \right), \quad (7)$$

then

$$\mathcal{E}(a + \lambda \delta) \leq \mathcal{E}(a) - \frac{\mu}{4} \lambda \|G(a)\|^2. \quad (8)$$

Proof. By the model equation, $Q_G(a, b) \delta = -G(a)$. Expanding $G(a + \lambda \delta)$ with the Taylor-defect assumption gives

$$G(a + \lambda \delta) = (1 - \lambda) G(a) + R_\lambda, \quad \|R_\lambda\| \leq \frac{L}{2} \lambda^2 \|\delta\|^2. \quad (9)$$

Since $\|\delta\| \leq K \|G(a)\|$, the defect is bounded by $\frac{L}{2} \lambda^2 K^2 \|G(a)\|^2$. The chosen bound on λ makes the quadratic error smaller than the linear decrease term. Squaring the residual and absorbing the second-order term yields the stated inequality. The constant $1/4$ is not optimized; it leaves room for non-normal finite-slope matrices and complex inner-product effects. $\square$

Remark 2.3. *This theorem explains the purpose of the modern Pandrosion probe machinery: equal-value avoidance, conditioning penalties, logarithmic normalization, and singularity filtering are not cosmetic. They are exactly the mechanisms needed to produce stable pairs (a, b) .*

3 Gated higher-order corrections

Finite-slope Halley corrections introduce a second displacement δ_2 through

$$Q_G(a, b)\delta_2 = -\frac{1}{2}H_G(\delta, \delta), \quad (10)$$

where $H_G(\delta, \delta)$ is estimated by symmetric finite-slope probes. The corrected direction is

$$\delta_H = \delta + \omega\delta_2, \quad (11)$$

with $0 \leq \omega \leq 1$.

Proposition 3.1 (Descent-preserving gate). *Assume the previous theorem holds for δ with a strict decrease margin $m > 0$. If the gate satisfies*

$$\omega \|\delta_2\| \leq c \lambda \|\delta\| \quad (12)$$

for a sufficiently small constant $c = c(L, K, m)$, then the Halley-corrected step is also energy-decreasing.

Proof. The corrected residual is the raw residual plus a perturbation controlled by local Lipschitz constants. The strict margin m absorbs perturbations of order $\omega \|\delta_2\|$. This is precisely the reason for a residual/condition/log-energy gate in practical engines. $\square$

4 Conjectures

Conjecture 4.1 (Pandrosion geometric descent conjecture). *For a regular isolated zero x^* of G , there exists a neighborhood V and a probe-selection rule using equal-value avoidance, logarithmic scale control, and singularity filtering such that all iterates starting in V satisfy*

$$\mathcal{E}(x_{k+1}) < \mathcal{E}(x_k) \quad (13)$$

until convergence.

Conjecture 4.2 (Global descent after dynamic normalization). *Let G be a polynomial residual map with finitely many regular zeros. After projective normalization and chart switching, each non-singular Pandrosion trajectory admits a subsequence of stable probe pairs for which the residual energy decreases monotonically outside a finite set of chart transitions.*

5 Implications for engine design

The theorem suggests a precise design rule. A Pandrosion engine should not maximize step size first; it should first maximize the probability that the selected pair (a, b) is stable. Once stability is obtained, line search and gated higher-order corrections provide descent. This explains why robust engines often outperform more aggressive ones on difficult systems.

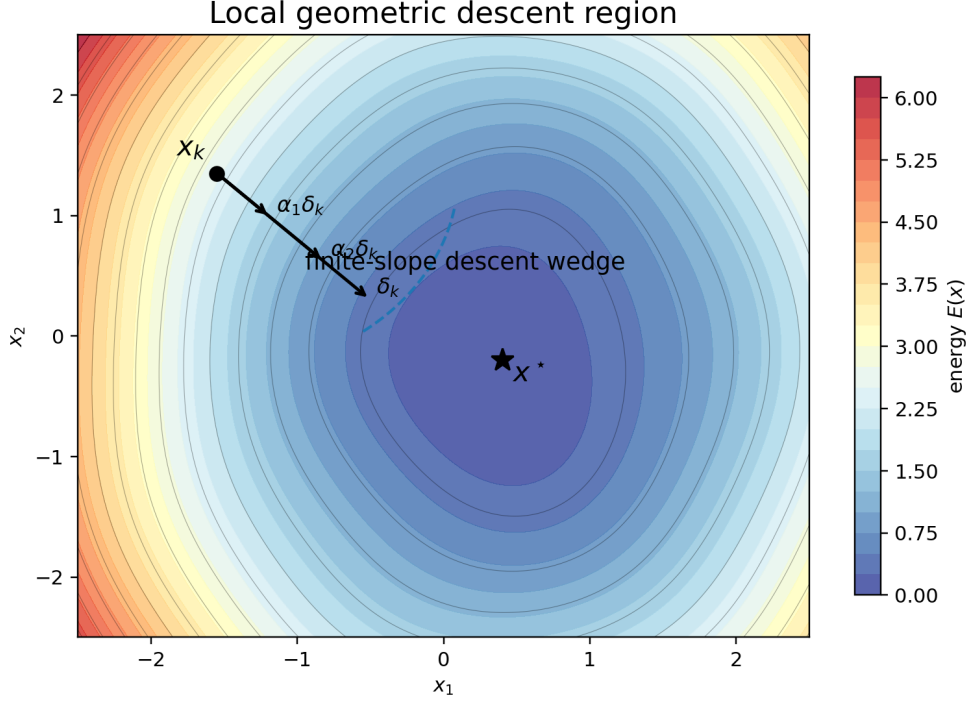


Figure 2: A supporting numerical-geometric schematic generated from the formal objects of the paper.

6 Outlook

The main purpose of this note is to isolate a precise mathematical direction. The next step is to attach each conjecture to a benchmarkable engine: finite-slope descent tests for the first paper, chart-selection tests for the second, and probe-path tests for the third. A successful theory should explain not only convergence, but also why the equal-value poles, logarithmic normalization, and projective charts observed in prior Pandrosion experiments are structurally necessary.

References

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